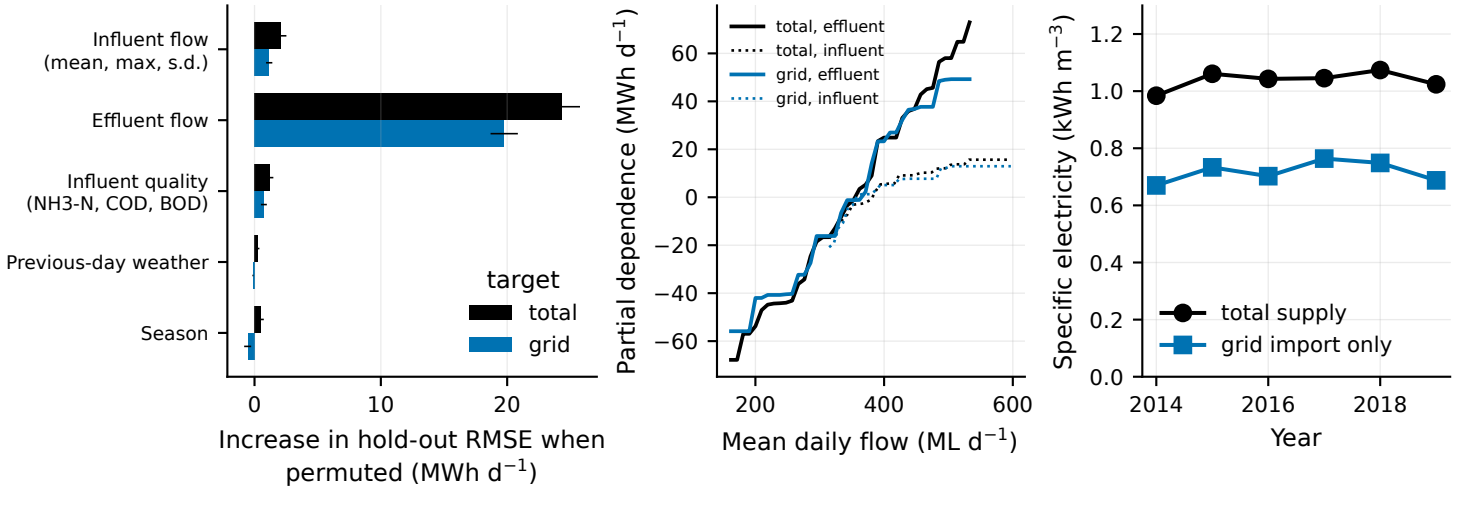
(a) permutation importance
Influent flow
(mean, max, s.d.)
Effluent flow
Influent quality
(NH3-N, COD, BOD)
Previous-day weather
Season
target
total
grid
0
10
20
Increase in hold-out RMSE when
permuted (MWh d^{-1})
(b) response to flow
total, effluent
total, influent
grid, effluent
grid, influent
Partial dependence (MWh d^{-1})
60
40
20
0
−20
−40
−60
200
400
600
Mean daily flow (ML d^{-1})
(c) specific electricity use
Specific electricity (kWh m^{-3})
1.2
1.0
0.8
0.6
0.4
0.2
0.0
total supply
grid import only
2014
2016
2018
Year